Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/08/2008
PI Name:	Liu, Feng-chuan Ph.D.		
Project Title:	Critical Thermal Transport in a Cross-Over Range from 3D to 2D behavior		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	10/01/1998	**End Date:**	09/30/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per D. Strayer (1/07) Note - Strayer says changed to ground in FY 2004		
Key Personnel Changes/Previous PI:	liu, feng-chuan		
COI Name (Institution):	Liu, Yuanming (JPL) Ahlers, Guenter (UCSB)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This task is part of the Boundary Effects on the Superfluid Tansition (BEST) Experiment previously scheduled to fly on the M2 MIssion of the Low Temperature Microgravity Physics Facility (LTMPF) on the International Space Station. BEST Experiment was returned to ground investigation in 2002 per NASA HQ direction as part of LTMPF funding reduction and re-plan. The science investigation studies the effects of a solid bounday on a fluid system near its critical transition, or specifically the superfluid transition of helium-4. The investigation consists of both measurement that can be done in the ground laboratory and measurement that can only be achieved in a microgravity environment in space. To perform systematic measurement on the thermal conductivity of helium-4 near its superfluid transition either as a 3-dimentional sample or in a 2-dimensional confinement. Comparing the results between the two boundary conditions will give tell us the effects and dimentionality and solid boundary on the superfluid transition. Confinement of different sizes willalso enabe us to test the hypothesis of dynamic finite size scaling theory. The task address two fundamental questions in physics: how does a solid boundary affect the critical phenomena in a fluid system, and does the finite size scaling work for dyanmic behaviors. Neither questions are properly understaood at this point because the investigation involves several conditions at the same time: a well defined confinement media, sub-nano-kelvin temperature and thermal contraol, as well as microgravity environment . With the microgravity environment on ISS and the stable low temparture environment provided by LTMPF, we have all the conditions to adress these important issues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Nature manifests itself differently for physical systems in three-dimension (3D), as compared to two-dimension (2D) or one-dimension (1D). While a system of fixed dimensionality may be well understood, the transition behavior from a 3D system to a 2D system is not well understood. A unique system to sudy the 3D to 2D transitional bahavior is helium-4 in presence of well defined solid boundary confinement and near the bulk superfluid transition (the lambda transition). As the lambda transition temperature is approached from the superfluid side, the corelation length diverges from well below, to comparable to, to well above the characterristic length of confinement in one direction, which makes the system behavior to change from 3D, to in-between 3D and 2D, to 2D. To accomplish this study, we have developed a new confinement media with well defined parallel-plate geometry and of different characteristic confinement lengths. In collaboration with Collimated Holes, Inc. and with Hamamatsu Corp, we have developed rectangular cross-sectional glass capillary arrays (GCA) with slot sizes from 1micron by 10micron, to 5micron by 50micron. Samples of various sizes should be studied in order to retrive the scaling behavior. In a ground laboratory, hydrostatic pressure limits the sharpness of the lambda transition, such that the sample with larger cross-section holes would have to be studied in microgravity. We completed construction of a fully functional, high-resolution low temperature laboratory at JPL, and conducted measurement of helium samples confined in GCAs of various confinement sizes.
Task Progress:	As part of the BEST effort, we have designed and developed a fully functional, state-of-the-art low temperature laboratory at JPL with the capability of performing research with nano Kelvin temperature control and precision testing. To enable the study of 3D to 2D transitional behavior, we collaborated with Collimated Holes and with Hamamatsu Corp. and developed new confinement media in form of Glass Capillary Arrays (GCA) with rectangular slots of 1 micron by 10 micron, and of 5micron by 50 micron. Much of the early effort of this project was devoted to developing technology necessary for the construction of a flight instrument. Examples include sample cell holders, pressure gauges and control, cryogenic valves, and high-resolution thermometry. Most of this development has benefited ground-based research. After the de-scoping of BEST we concentrated on ground-portion of the planned BEST measurements on the thermal conductivity of helium in confined geometries. We completed ground measurements of the conductivity in plates with channels of 1x10 micron cross section, and simultaneously to bulk helium. The bulk measurements served to provide the bulk lambda-point as a fixed point on the temperature scale. These results have been published very recently in the Journal of Low Temperature Physics. To our great surprise the results show that the finite-size scaling-function for this geometry does not differ measurably from the scaling function for cylindrical confinement. This differs qualitatively from what had been found for thermodynamic properties in one-dimensional and two-dimensional confinement.
Bibliography Type:	Description: (Last Updated: 04/08/2008)
Articles in Peer-reviewed Journals	Murphy D, Genio E, Ahlers G, Liu F, Liu Y. "Finite-size scaling and universality of the thermal resistivity of liquid 4He near Tlambda." Phys Rev Lett. 2003 Jan 17;90(2):025301. PMID: 12570552 http://dx.doi.org/10.1103/PhysRevLett.90.025301 , Jan-2003
Articles in Peer-reviewed Journals	Genio E, Murphy D, Ahlers G, Liu F-C, Liu, Y-M. "Scaling and universality of the thermal conductivity of liquid 4He near the superfluid trantransition and in restricted geometries." Advances in Space Research, 2005;35(1):87-91. http://dx.doi.org/10.1016/j.asr.2003.03.046 , Jan-2005
Articles in Peer-reviewed Journals	Jerebets S, Liu Y, Liu F, Ahlers G. "Finite-size effects on the thermal resistivity of 4He near the superfluid transition." Journal of Low Temperature Physics. 2007 Mar;146(5-6):471-83. http://dx.doi.org/10.1007/s10909-006-9278-5 , Mar-2007